Supplementary methods

Joint Learning of Safety-Relevant Anatomy from Heterogeneous Supervision in Laparoscopic Cholecystectomy

S1. Training configuration

Table 1 describes the full joint model used for joint-model results and fixed-model test-phase analysis. The joint/CBD-only and fluorescence comparisons reuse the full joint model's optimizer, learning-rate schedule, batch size and LoRA settings; their checkpoint criterion is specified below. CBD-only sampling and phase controls retain these settings, with 240 warm-up updates and cosine decay to zero at update 4,000 (S4–S5). Source-combination runs retain the full joint recipe, with source weights recomputed for retained datasets. Augmentations comprise horizontal flips, color jitter, Gaussian blur, rotations and perspective warping.

Table 1 Optimization settings for the full joint SAM3 model.

Setting	Value
Optimizer	AdamW; $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$
Weight decay	0.01
Learning rate	Peak 10^{-4}
Schedule	Linear warm-up for 240 updates, then cosine decay to zero
Duration	12,000 optimizer updates; no early stopping
Batch size	8 images per update; gradient accumulation 1
Precision	BF16 mixed precision
Gradient clipping	Maximum gradient norm 1.0
Random seed	42
Input	Aspect-ratio-preserving letterboxing to 1008×1008
Validation/checkpoints	Every 250 updates

Low-rank adaptation.

Only LoRA parameters are optimized; pretrained SAM3 parameters remain frozen. The adapter rank is $r = 16$, scaling parameter $\alpha = 32$ (hence $\alpha/r = 2$), and dropout 0.1. Adapters are applied to the vision encoder, DETR encoder and decoder, and segmentation head. They target attention projections, transformer feed-forward linear layers and the mask-prediction MLP. The text encoder, geometry encoder and decoder text cross-attention remain frozen without LoRA. The run contains 4,874,240 trainable parameters, approximately 0.577% of the model (rounded to 0.6% in the main text).

Checkpoint selection.

Validation and checkpoint saving occur every 250 updates. All-target models select by validation NGZ mask AP50. For the full joint model, this is the unweighted mean across the iBSAFE and 2C-ICG validation sets. Endoscapes does not enter this criterion because it has no NGZ targets. Official test data are not used for selection. CBD-only models select by validation CBD box AP50. All six joint/CBD-only

configurations use 2,162 frames and one 12,000-update run each. Both fluorescence-comparison models complete 12,000 updates using all 6,577 iBSAFE training frames; the two-source model additionally uses all 1,176 Endoscapes training frames.

Line-supervision band.

The rasterized reference curve is dilated by 5×5 max pooling, creating a two-pixel band on each side at the predicted-mask resolution. The positive term samples 128 positions along the annotated curve; the negative term acts outside the dilated band. The curve is not used as a dense focal/Dice mask target. This band is independent of the five-pixel evaluation tolerance defined in S3.

S2. Source sampling

For a retained source d with N_d unique optimization frames after validation exclusion,

$$P(d) = \frac{\sqrt{N_d}}{\sum_j \sqrt{N_j}}.$$

The iBSAFE source combines keyframes and uniform frames with shared frames counted once. Table 2 gives the source counts and their square-root sampling shares. A sampler epoch contains 8,085 draws, the sum of the source-frame counts. Frames are shuffled within each dataset, and smaller datasets cycle through fresh permutations to fill their quota. These source weights govern sampling, not loss coefficients. When a dataset is omitted, the weights are recomputed over the retained sources.

Table 2 Full joint-model sampling after validation exclusion. Shares are calculated from the source counts and rounded.

Source	N_d	Share (%)
iBSAFE, deduplicated key + uniform	6,577	60.70
Endoscapes	1,176	25.67
2C-ICG	332	13.64

S3. Landmark-line extraction and scoring

For each RS or S4b prediction, mask logits are mapped back to the native cropped-image coordinates and binarized at logit > 0, equivalent to foreground probability > 0.5. Class-wise box non-maximum suppression uses IoU 0.5 and retains at most 100 detections. The primary results use detection confidence ≥ 0.5. Retained masks are united separately for each landmark class, then converted to thin binary curves by Zhang–Suen skeletonization. No spline fitting, largest-component selection or branch pruning is applied. Ground-truth polylines are rasterized as one-pixel-wide lines.

Distances are measured on the native grid and scaled to a longest image edge of 1008 pixels. At the primary tolerance $\delta = 5$, precision is the fraction of predicted curve

Table 3 CBD AP50 (%). First three rows provide means and standard deviations across five training-subset draws. The fourth row is a single-run full-data result.

Training policy	Frames	Key test	Uniform test
Key	2,162	78.3 ± 1.4	72.4 ± 1.8
Uniform	2,162	80.2 ± 1.2	76.8 ± 1.3
Combined	2,162	82.0 ± 1.0	75.3 ± 1.5
Combined, full	6,577	83.0	78.0

pixels within δ of the reference, and recall is the fraction of reference pixels within δ of the prediction. Matched predicted/reference pixel counts and their respective totals are pooled across eligible frames before computing $F1_\delta = 2P_\delta R_\delta/(P_\delta + R_\delta)$. Exhaustively annotated negative frames are retained. RS uses 616 uniform test frames; S4b uses the 199 uniform pre-HCT frames.

S4. Matched-budget frame-sampling controls

We compare CBD-only SAM3 models using 2,162 frames from 85 iBSAFE videos per arm. Within each video × phase stratum, the keyframe and uniform arms receive the same frame count; the combined arm receives that same total, half keyframes and half uniform. This is the maximum keyframe count after validation exclusion and subset selection. We repeat subset selection five times at a fixed training seed, with 4,000 updates per fit and checkpoint selection by validation CBD box AP50. The setups match phase composition, supervision, training protocol and video coverage. Fluorescence evidence is not balanced: CBD fluorescence-point presence is recorded descriptively and does not enter frame selection. Point absence on keyframes cannot reliably indicate weak fluorescence evidence. A fourth model uses all 6,577 combined optimization frames. All models use the same official keyframe and uniform tests; no other dataset enters training or validation.

Uniform training improves uniform-test CBD AP50 by 4.4 points over keyframe training; combined sampling gives the highest matched-budget mean on keyframe tests (Table 3).

S5. Training-phase controls

We compare CBD-only SAM3 models on uniformly sampled iBSAFE frames, training on pre-HCT, HCT or an equal mixture. Each setup uses 866 frames from the same 79 videos, with identical per-video totals; the mixture contains 433 frames per phase. Five subset draws use a fixed training seed and 4,000 updates per fit, selecting checkpoints by validation CBD box AP50. A fourth model uses all 3,899 optimization frames from 86 videos. Tests contain 199 pre-HCT and 417 HCT frames.

Mixed-phase training gives the highest mean pre-HCT AP50, whereas HCT-only training performs best on HCT (Table 4). Thus, matching training and test phase does not consistently help, and mixing does not improve both endpoints. The full-data model performs best on both tests, but additionally changes frame count, video coverage and phase proportions.

Table 4 CBD AP50 (%) for SAM3 phase controls on iBSAFE-u. Matched-budget rows give mean ± sample SD over five subset draws; the full-data row is one run.

Training phase	Training frames	pre-HCT test	HCT test
pre-HCT	866	67.87 ± 2.72	68.69 ± 1.62
HCT	866	69.22 ± 4.78	75.60 ± 0.23
Both, matched budget	866	71.13 ± 1.44	73.93 ± 3.29
Both, full data	3,899	75.16	82.12